

Atmospheric Correction

G2GS

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Table of Contents

1	INTRODUCTION.....	4
1.1	Purpose of This Document.....	4
1.2	Who Should Use This Document?	4
1.3	Inside Each Section.....	4
1.4	Related Documents	4
1.5	Revision History.....	4
1.6	List of Acronyms	5
2	ALGORITHM DESCRIPTION.....	6
2.1	Algorithm Overview	6
2.1.1	Product Generated.....	8
2.1.2	GOCI-II Characteristics	8
2.2	Processing Outline	10
2.3	Algorithm Input	11
2.3.1	Primary Sensor Data.....	11
2.3.2	Ancillary Data.....	11
2.4	Physics of the Problem.....	12
2.5	Mathematical Description	12
2.5.1	Structure and Interpolation of LUT data	12
2.5.2	GOCI-II Atmospheric Correction (G2AC)	13
2.6	Algorithm Product Output	22
2.6.1	Output	22
2.6.2	Flag	23
3	TEST DATA SETS AND OUTPUT.....	26
3.1	Simulated / Proxy Input Datasets	26
3.1.1	In situ radiometric data measured from shipboard	26
3.1.2	AERONET-OC radiometric data	26
3.1.3	Simulation dataset.....	28
3.2	Output from Simulated Datasets	28
3.2.1	Accuracy Estimates.....	28
4	PRACTICAL CONSIDERATIONS.....	29
4.1	Numerical Computation Considerations.....	29
4.2	Algorithm Validations.....	29
5	ASSUMPTIONS AND LIMITATIONS.....	30
5.1	Performance	30
5.2	Assumed Sensor Performance.....	30
6	REFERENCES.....	31
7	APPENDIX.....	38

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1 INTRODUCTION

1.1 Purpose of This Document

This ATBD document describes the algorithm background for the GOCI-II atmospheric correction (G2AC) that produces Rayleigh-corrected reflectance (ρ_c), remote-sensing reflectance (R_{rs}), and normalized-water-leaving radiance (nL_w). This algorithm was developed based on the GOCI atmospheric correction algorithm (Ahn et al., 2012; 2015; 2016; 2018) that modification of SeaWiFS standard atmospheric correction algorithm (Gordon and Wang 1994; Bailey et al., 2010) to enhance the accuracy regarding aerosol model selection from candidates and turbid water NIR correction. Note that the current G2AC is an initial phase of the development cycle. The algorithm will be updated with further considering additional bands: 380, 620, and 709 nm wavelengths.

1.2 Who Should Use This Document?

This document was created for the researchers who would like to know about the theoretical background and implementation of G2AC algorithm.

1.3 Inside Each Section

This document is composed as follows.

- **Introduction:** brief description on GOCI-II system and the atmospheric correction products
- **Algorithm Description:** input/output to the G2AC algorithm, physical principle, and description
- **Test Datasets and Outputs:** data and results for the G2AC test
- **Practical Considerations:** provide the guideline about the algorithm maturity, S/W implementation information, data quality assessment, exception handling, which will be helpful for the people who regularly maintain and operate the G2AC algorithm and software.
- **Assumptions and Limitations:** suggests the restriction in application, the uncertainty of input data, and the potential solution based on the performance of the system assumed at the development stage.

1.4 Related Documents

This document is related to the following GOCI-II ATBD document.

- GOCI-II ATBD for Vicarious Calibration

The documents can be found in the KIOST website (the website location is subject to change).

1.5 Revision History

Revision	Date	Revision Description	Page Affected
0.1	2017-09-01	Initial Issue	
0.2	2017-10-07	ATBD writing instruction is added and updated	all
0.21	2017-10-30	Translated into English	

Revision	Date	Revision Description	Page Affected
0.25	2020-08-11	Updated with considering current development	all
0.3	2021-05-14	Some minor typos are corrected	all

1.6 List of Acronyms

Acronyms used in this document is listed in the Table 1.

Table 1. Acronyms table

LG	[Gain Mode] Low Gain
HG	[Gain Mode] High Gain
GOCI-II	Geostationary Ocean Color Imager-II
GK2	Geostationary KOMPSAT 2
G2AC	GOCI-II atmospheric correction algorithm
INR	Image Navigation and Registration
DCS	Data Correction Subsystem (KIOST)
DPS	(=DCS) Data Pre-processing Subsystem (KARI)
CADU	Channel Access Data Unit
DAMD	Diffuser Aging Monitoring Device
DK	Dark
ST	Star
SD	Solar Diffuser
SCN	Scene

2 ALGORITHM DESCRIPTION

2.1 Algorithm Overview

The GOCI-II remotely senses ocean color information from geostationary orbit (altitude: ~36,000km) compare to sun-synchronous orbit sensors (altitude: ~700km). Extracting the water-leaving radiance from total radiance recorded at the top-of-atmosphere (TOA), however, contains more than 90% of atmospheric path radiance resulted from multiple-scattering of light by air molecules and aerosols including their interactions (Figure 1 and 2) (Gordon, 1998; Eplee et al., 2001; Wang and Gordon, 2002; Franz et al., 2007). Therefore, the atmospheric correction is an essential process for the ocean color remote sensing including the GOCI-II mission. The atmospheric correction is a sensitive process due to the relatively low ratio of water radiance to atmospheric radiance, i.e., 1% error in estimation of the atmospheric radiance would cause more than 10% error in the ocean radiance estimation (Eplee et al., 2001).

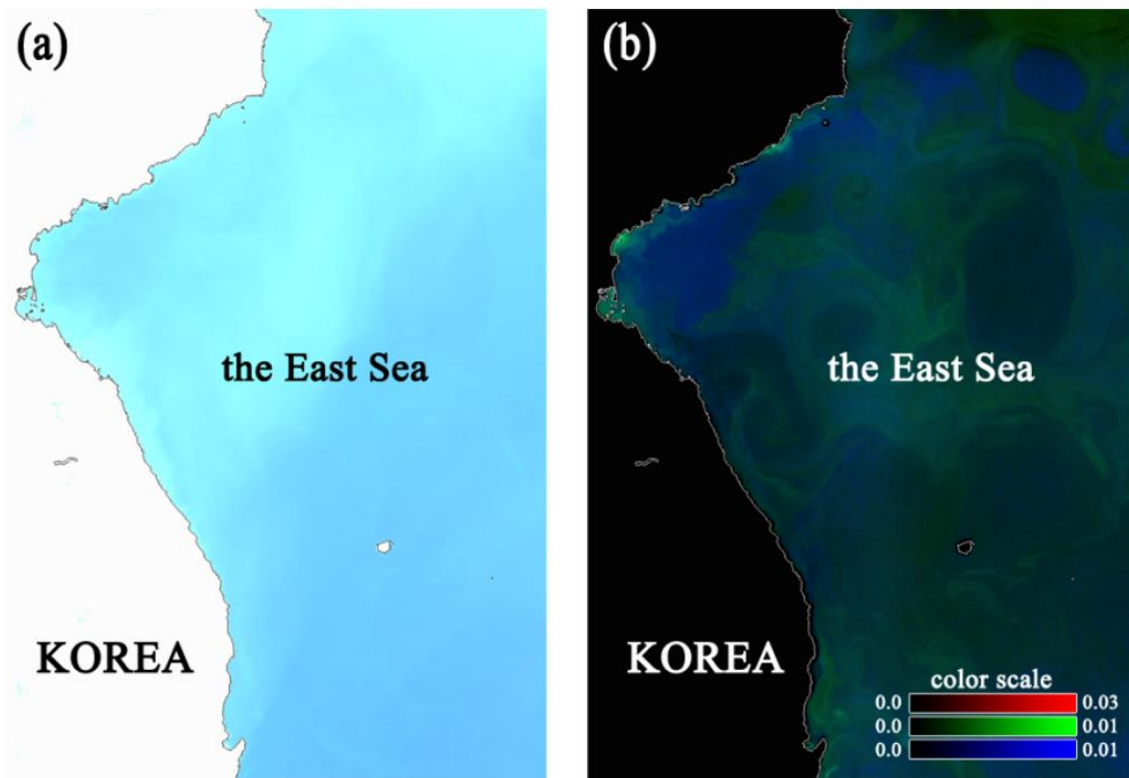


Figure 1. GOCI RGB (660 nm, 555 nm, and 443 nm, respectively) composite images of TOA reflectance in same color scale before the atmospheric correction (a), and water reflectance after the atmospheric correction (b) over the East Sea. The example imagery have been obtained in 2011/04/05 03:16 UTC.

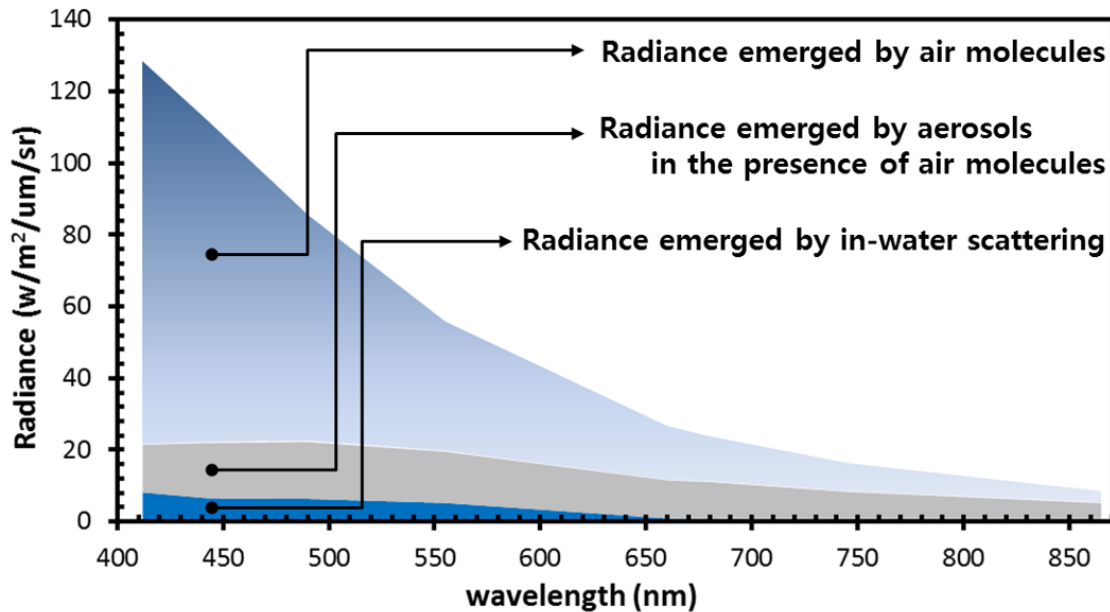


Figure 2. Spectra of TOA radiance contributors at 2011/04/05 03:16 UTC, in the middle of the East Sea, from GOCI observation. In the observation, water-leaving radiance to atmospheric path radiance is less than ~7% in blue bands.

Earlier atmospheric correction for the ocean color remote sensing considered only the open seas. Thus, it assumed that the water-leaving radiance is negligible in the near-infrared (NIR) spectral range and is called a black pixel assumption (BPA). The BPA has made possible to know the atmospheric radiance in NIR wavelengths over the ocean. The initial BPA-based atmospheric correction algorithm was first developed for the CZCS data processing by Gordon (1978, 1983). This algorithm is based on the single-scattering approximation for aerosol particles. For low aerosol optical thickness (AOT) conditions (i.e., $AOT < 0.1$), this single-scattering-based atmospheric correction had worked acceptably. Later, an improved atmospheric correction algorithm considering a multiple-scattering for various aerosol type for the operational data processing of the Sea-viewing Wide Field-of-view Sensor (SeaWiFS) was developed by Gordon and Wang (1994) and Wang and Gordon (1994). This algorithm is an update version of the CZCS approach regarding the aerosol correction scheme. Thus the process first converted the multiple-scattering into the single-scattering to select aerosol models and extrapolate aerosol's single-scattering reflectance in NIR band to VIS bands. After the aerosol model selection, the aerosol single-scattering reflectance in the VIS bands are converted back into the VIS bands. Later, the SeaWiFS algorithm was slightly modified for the Ocean Color and Temperature Scanner (OCTS) by Fukushima et al. (1998). The OCTS algorithm replaced the single-scattering reflectance to the AOT to simplify the process, and subsequently enhanced the aerosol model selection scheme with considering weighted-average technique. More recently, a new multiple-scattering atmospheric correction algorithm was developed by Antoine and Morel (1999) for the Medium Resolution Imaging Spectrometer (MERIS) data processing. The MERIS algorithm also converts the multiple-scattering reflectance to the AOT similar to the OCTS approach. However, it selects the appropriate aerosol models in the multiple-scattering space. The performance of those three primary algorithms was evaluated by the International Ocean-Colour Coordinating Group (IOCCG) report No. 10 (2010), and it showed all of three algorithms were working well within the acceptable error range (– less than 5% error in the blue band for general sun-sensor geometries).

The G2AC algorithm is also developed based on the SeaWiFS and CZCS atmospheric correction approach with partial improvements with focusing on aerosol correction scheme, turbid water NIR correction, and vicarious calibration (Ahn et al., 2012, 2015, 2016). The aerosol correction scheme in the G2AC algorithm is developed to overcome some assumptions and limitations associated with existing methods that rely on either single-scattering reflectance or aerosol optical depth. The new aerosol correction scheme is simpler and methodologically

more straightforward. Moreover, it estimates the aerosol multiple-scattering reflectance at the visible wavelengths more accurately by using the spectral relationships in the aerosol multiple-scattering reflectance between different wavelengths (called SRAMS).

To extend the ocean color remote sensing to the coastal environment monitoring, atmospheric correction over turbid water (where the BPA is not valid) has become important. Several studies have addressed the atmospheric correction over the turbid waters (Hu et al., 2000; Ruddick et al., 2000; Siegel et al., 2000; Wang and Shi, 2007; Stumpf et al., 2003; Bailey et al., 2010; Wang et al., 2012; Jiang and Wang, 2014). Especially, the GOCI-II observation coverage contains highly turbid water areas such as the Yellow Sea, the East China Sea, and the coastal areas of the Korean peninsula. Thus, an implementation of the G2AC algorithm can be applied to both the turbid waters and the clear waters. This study also focuses on an improvement of the turbid water (i.e., inorganic particles are dominant) atmospheric correction for GOCI-II target area. A strong empirical relationship between water reflectance derived by satellite at red and two NIR bands can be established in a wide range of water turbidity. The G2AC process iteratively separates water reflectance and multiple-scattering reflectance of aerosols by the empirical relationship model.

2.1.1 Product Generated

This section describes the purpose of the G2AC algorithm, and what is the advantage of GOCI-II for the G2AC process.

The atmospheric correction is required to retrieve the surface radiance from remotely sensed TOA-radiances by removing the atmospheric effects. In a typical open-ocean region of oligotrophic waters, the upwelling radiance emerging out of the water surface contributes ~ 10% to the radiance at the TOA. [p1] Therefore, it is crucial to retrieve this small portion of the water reflectance at the sea surface by removing the major portion of atmospheric (molecules and aerosols) contributions and specular reflection at the sea surface.

The algorithm generates remote-sensing reflectance of water at the surface (R_{rs}), Rayleigh corrected reflectance from the TOA (ρ_c), and other intermediate parameters related to the atmospheric correction process.

The G2AC products must have the same temporal and spatial resolution with the GOCI-II L1B data.

Table 2. Product requirements of G2AC

Product level	Spatial resolution	Temporal resolution	Acc. for case-1 waters	Acc. for case-2 waters
Level-2	300 m	1 hour	80 %	70 %
Level-3	1 km	1 day	-	-

2.1.2 GOCI-II Characteristics

The Second Geostationary Ocean Color Imager (GOCI-II) is one of the two payloads on the Geostationary Korea Multi-Purpose Satellite-2B (GK2B) operated by Korea Ocean Satellite Center (KOSC) of Korea Institute of Ocean Science and Technology (KIOST) (Figure 3). From the geostationary orbit, GOCI-II provides the regional synoptic perspectives of coastal and open ocean phenomena around the Northeast Asia Seas and the earth full disk. It can take daytime images with unprecedented temporal resolution (10 times a day from 08:15 to 17:15, local time GMT+9) at one near UV band (380 nm), eight visible bands (412, 443, 490, 510, 555, 620, 660, and 680nm), and three NIR bands (709, 745, and 865nm) and a moderate spatial resolution (~300m at the scene center). The spectrally expanded wavelengths of GOCI-II observation makes more advantages compare to GOCI mission for absorbing aerosols (380 nm) and turbid water NIR (620, 709 nm) correction.

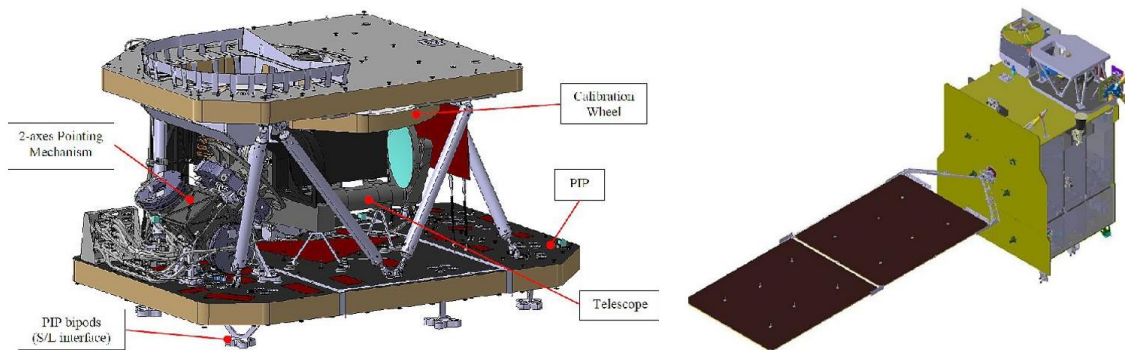


Figure 3. The main unit of the GOCI-II without multi-layer insulation protection (left) and GEO-KOMPSAT-2B (GK2B) spacecraft (right). GOCI-II is one of two payloads onboard GK2B. (image credit: KARI)

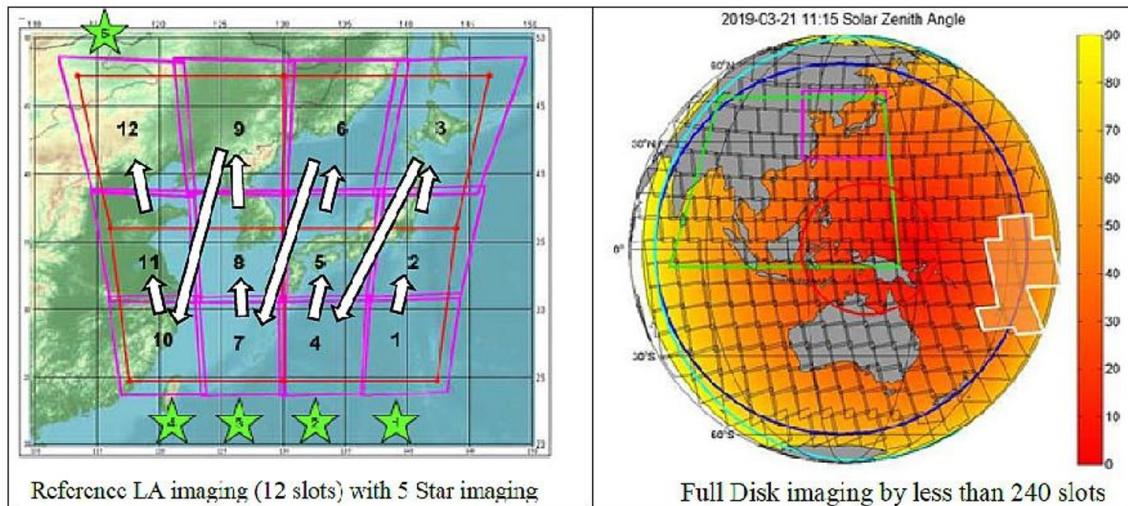


Figure 4. The GOCI-II observation area for two imaging modes: Northeast Asia (left) and Full Disk acquisition (right).

Table 3. Characteristics of the spectral bands of GOCI

Band	Band Center (nm)	Bandwidth (nm)	Nominal Radiance	Maximum Ocean Radiance	Saturation (Threshold) Radiance	Maximum Cloud Radiance	SNR @ Nominal Radiance
B1	380	20	93	139.5	143.1	634.4	998
B2	412	20	100	150	152	601.6	1050
B3	443	20	92.5	145.8	148	679.1	1145
B4	490	20	72.2	115.5	116	682.1	1228
B5	510	20	55.3	85.2	122	665.3	1124
B6	555	20	55.3	85.2	87	649.7	1124
B7	620	20	40.3	67.8	70.5	616.5	1080

Band	Band Center (nm)	Bandwidth (nm)	Nominal Radiance	Maximum Ocean Radiance	Saturation (Threshold) Radiance	Maximum Cloud Radiance	SNR @ Nominal Radiance
B8	660	20	32	58.3	61	589	1060
B9	680	10	27.1	46.2	47	549.3	914
B10	709	10	27.7	50.6	51.5	450	914
B11	745	20	17.7	33	33	429.8	903
B12	865	40	12	23.4	24	343.8	788
B13	643.5 (PAN)	483	-	-	-	-	-

Table 4. Specification of the GOCI spectral observation

Band	Band Center (nm)	Primary Usage
B1	380	CDOM (Chromophoric Dissolved Organic Matter), absorbing aerosol correction
B2	412	CDOM, chlorophyll
B3	443	Chlorophyll absorption maximum
B4	490	Chlorophyll and other pigments
B5	510	Chlorophyll, absorbing aerosol in oceanic waters
B6	555	Turbidity, suspended sediment
B7	620	Phytoplankton species detection
B8	660	Baseline of fluorescence signal, Chlorophyll, suspended sediment
B9	680	Fluorescence signal
B10	709	Fluorescence base signal, atmospheric correction, suspended sediment
B11	745	Atmospheric correction, vegetation index
B12	865	Atmospheric correction, aerosol optical depth
B13	Wideband	Star Imaging for the INR (Image Navigation and Registration) performance

2.2 Processing Outline

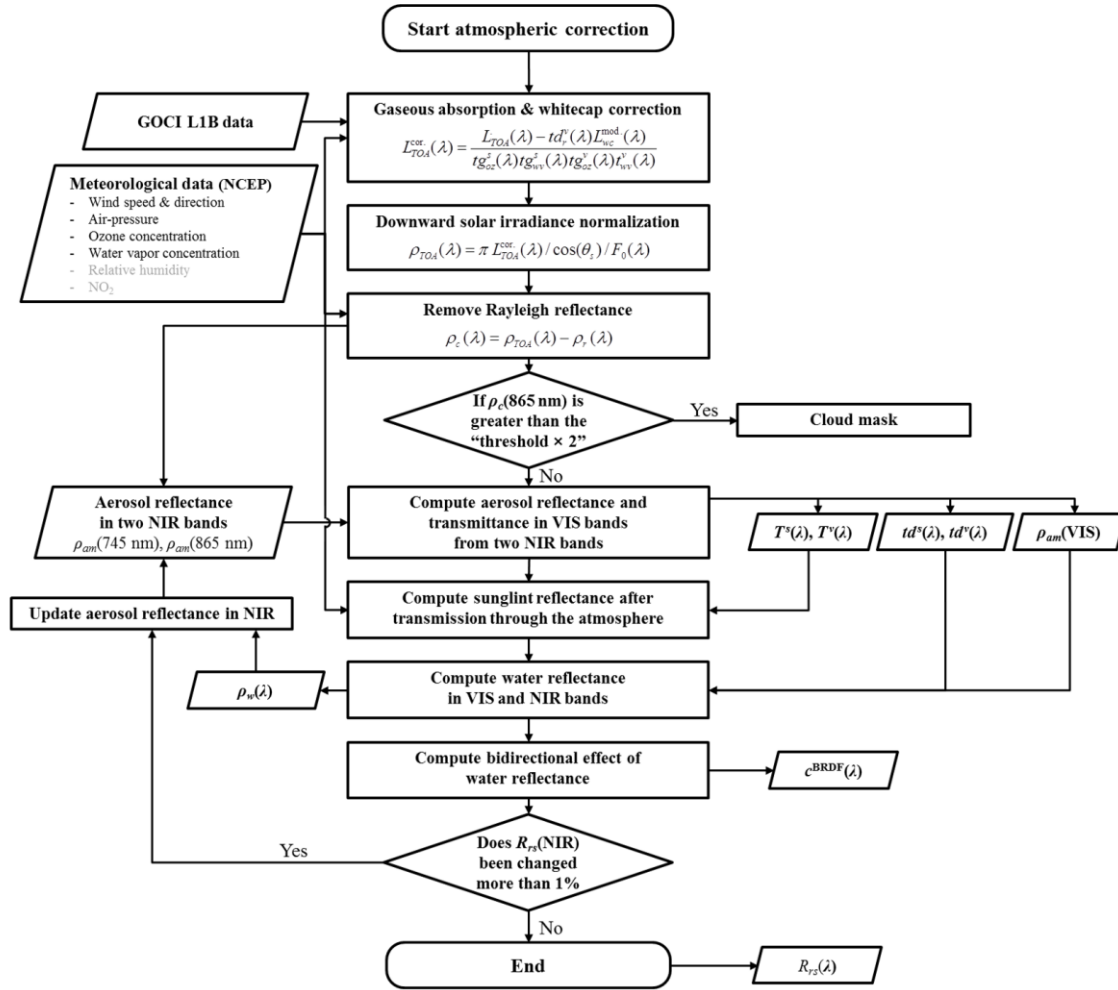


Figure 5. Flowchart of G2AC

2.3 Algorithm Input

This section describes about the input data for the G2AC algorithm. As input data, GOCI-II basically takes the TOA radiance (GOCI-II level-1B) with scan-geometries after the radiometric and geometric corrections. Ancillary meteorological data that are wind field, air pressure, relative humidity, ozone, water vapor, nitrogen dioxide downloaded from ancillary data server can be used to correct the effect of surface roughness, Rayleigh optical thickness, aerosols' optical characteristics, and trace-gas absorption, respectively. The G2AC process also uses look-up tables that stores both forward and backward model of radiometric transfer for each specific scan-geometries and meteorological conditions.

2.3.1 Primary Sensor Data

GOCI-II Level-1B L_{TOA} and related scan-geometry products as an input.

2.3.2 Ancillary Data

G2AC uses ancillary meteorological data that are wind field, air pressure, relative humidity, ozone, water vapor, nitrogen dioxide. The NCEP (www.ncep.noaa.gov) meteorological prediction data can be first applied in a near-real time for the first operational data processing, then ECMWF-ERA5 (www.ecmwf.int) re-analysis data is applied as a scientific quality data processing two or three months later. If the meteorological data is not available for temporarily,

it can be substituted by a climatology data stored as a look-up table. G2AC process mark this issue in both flag and metadata, because G2AC products generated with the climatology data can be less accurate.

2.4 Physics of the Problem

An atmospheric correction algorithm aims to retrieve the water reflectance (r_w) or remote-sensing reflectance (R_{rs}) from the total radiance recorded by a satellite ocean color sensor at the top-of-the-atmosphere (TOA) by removing atmospheric path radiance. Contributors of atmospheric path radiance at TOA level can be separately considered that are multiple scattering by air molecules (Rayleigh scattering) in the absence of aerosols (Figure 6 (a)), multiple scattering by aerosols in the absence of air molecules (Figure 6 (b)), and interactions between aerosols and molecules (Figure 6 (c)) with considering not only radiances between sea surface and satellite but also Fresnel-reflected atmospheric radiances at the air-sea interface. An atmospheric correction algorithm aims to retrieve the water-leaving reflectance or remote-sensing reflectance (Figure 6 (d)) from the total radiance recorded by a satellite ocean color sensor by removing atmospheric path radiance.

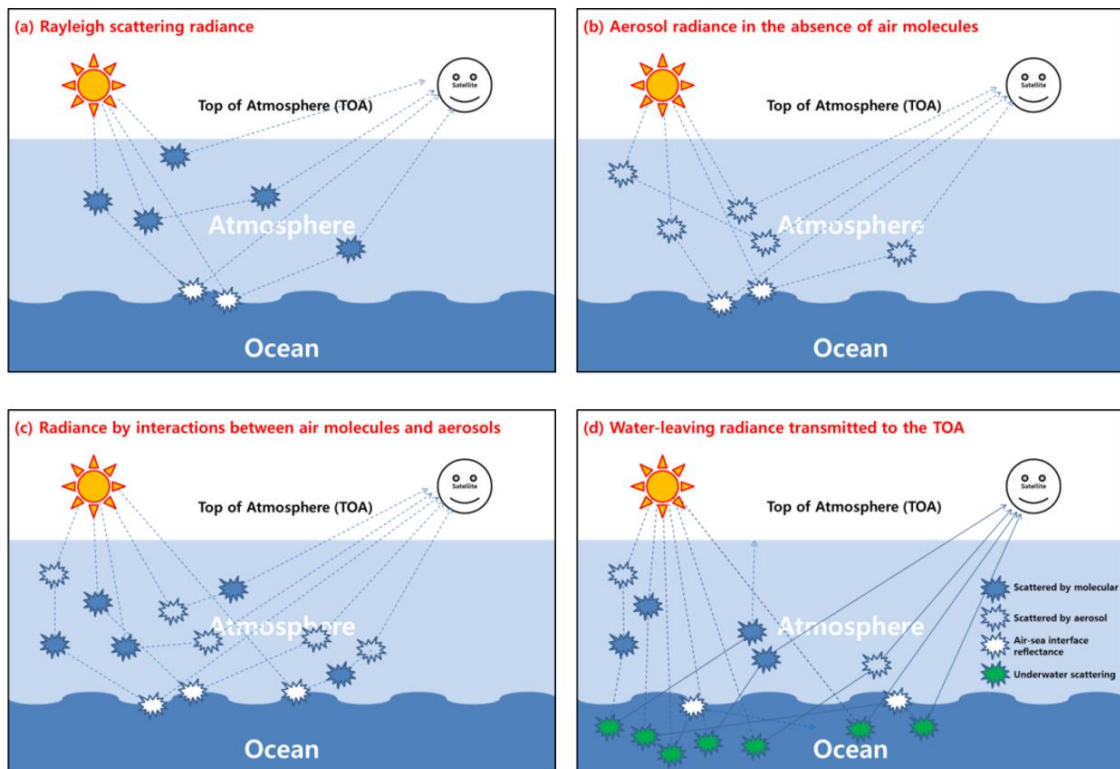


Figure 6. Contributors of the TOA radiance. (a) Multiply scattered radiance by air molecules (Rayleigh scattering), (b) multiply scattered radiance by aerosols, (c) radiance by interactions between air molecules and aerosols, (d) water-leaving radiance that has to be retrieved eventually.

2.5 Mathematical Description

2.5.1 Structure and Interpolation of LUT data

This section describes look-up tables (LUT) used by G2AC. In the G2GS process, LUT for various scan-geometries are stored to calculate Rayleigh reflectance, aerosol reflectance,

BRDF of water reflectance at the surface, and atmospheric transmittance by air molecules, aerosols, and trace gases.

The Rayleigh reflectance LUT has following structure,

Stored Data

- Rayleigh reflectance over ocean and land
- Rayleigh diffuse transmittance from the sun to the sea surface and from the sea surface to sensor.

Dimensions

- 12 spectral bands, 5 wind, and different scan geometries
- Data type: float (4 bytes)

The aerosol LUT has following structure,

- Total 13 aerosol models as a candidate: O99, M98, M95, M90, M80, M70, M50, C90, C70, C50, T90, T70, T50 (Shettle and Fenn, 1979).
- Coefficients for the 11 spectral relationships of aerosol reflectance model for different scan geometries
- Data type: float (4 bytes)

The f/Q BRDF LUT has following structure,

Stored Data

- f/Q correction factors for different scan geometries and IOPs
- Data type: float (4 bytes)

The $\mathfrak{R}_s$ BRDF LUT has following structure,

Stored Data

- Coefficients for downward Fresnel transmittance ($\mathfrak{R}_s$) correction model for 7 visible spectral bands, 5 wind speeds, and different scan-geometries
- Data type: float (4 bytes)

2.5.2 GOCI-II Atmospheric Correction (G2AC)

Table 5. Glossary of symbols

symbols	description
λ	wavelength
θ_s	solar zenith angle
θ_v	viewing (satellite) zenith angle
ϕ_{s-v}	azimuth angle difference between the sun and the sensor at the position
0+	just above the water
0+	just beneath the water
Ed	downward irradiance
Eu	upward irradiance

symbols	description
chl_a	chlorophyll-a
L_{TOA}	top-of-atmosphere radiance observed by satellite
L_{TOA}^{cor}	top-of-atmosphere radiance after the correction of white cap and gaseous absorption
$L_a(\lambda)$	multiple scattering radiance by aerosols in the absence of air molecules
$L_{ra}(\lambda)$	radiance by light interaction between air molecules and aerosols
L_r	multiple scattering radiance by air molecules (Rayleigh scattering) in the absence of aerosols
L_{wc}	whitecap radiance
L_w	water-leaving radiance
nL_w	normalized water-leaving radiance
L_{sea}^{0+}	total above water radiance leaving the water surface with including sea-surface reflected sky radiance
L_{sky}	Sky radiance at the surface
$L_b(\lambda)$	the residual radiance from light reflected by the ship's superstructure, microfoam, or fluctuated $f_{surf} \times L_{sky}$
R_b	L_b/Ed
R_{rs}	remote-sensing reflectance
R_{rs}^m	remote sensing reflectance before the bidirectional effect correction
ρ_{TOA}	top-of-atmosphere reflectance
ρ_w	water reflectance (reflectance at the top-of-atmosphere)
ρ_{wn}	normalized water reflectance (reflectance at the surface)
ρ_{wc}	whitecap reflectance
ρ_r	multiple scattering reflectance by air molecules (Rayleigh scattering) in the absence of aerosols
ρ_r^{1atm}	preliminary computed Rayleigh scattering reflectance for 1 atmospheric pressure
ρ_a	multiple scattering reflectance by aerosols in the absence of air molecules
ρ_{ra}	reflectance by light interaction between air molecules and aerosols
ρ_{as}	aerosol single-scattering reflectance
ρ_{am}	$\rho_a + \rho_{ra}$
ρ_{am}^{Mod}	theoretically computed $\rho_{am}(\lambda)$ for specific aerosol model
ρ_g	sun-glint reflectance that is spectrally independent
r_{rs}^{model}	the remote-sensing reflectance just below the surface
b_b	total backscattering coefficient
a	total absorption coefficient

symbols	description
$\mathfrak{R}$	Fresnel transmittance at the air-sea interface
f	ratio of underwater IOP term to under water reflectance
Q	underwater bidirectional effect term
C_{BF}	bidirectional effect coefficient (i.e., $\mathfrak{R} \cdot f / Q$)
ε	aerosol single-scattering reflectance ratio (i.e., single-scattering epsilon, SSE)
ε^{pre}	theoretical SSE value
ε^{ave}	averaged ε for candidate aerosol models
td^s	total downward diffuse transmittances including both the aerosols and the air molecules
td^v	total downward diffuse transmittances including both the aerosols and the air molecules
td_r^s	downward diffuse transmittance by air molecules
td_r^v	upward diffuse transmittance by air molecules from the sea surface to the sensor
tg^s	total downward direct transmittances including both the aerosols and the air molecules
tg^v	total upward direct transmittances including both the aerosols and the air molecules
t_{oz}^s	downward gaseous transmittance by ozone from the sun to the sea surface
t_{oz}^v	upward gaseous transmittance from the sea surface to the sensor
t_{ww}^s	downward gaseous transmittance by water vapor from the sun to the sea surface
t_{ww}^v	upward gaseous transmittance by water vapor from the sun to the sea surface
tdm^s	downward diffuse transmittance model established with assuming that the single scattering albedo multiply forward scattering probability is 0.95 for j^{th} aerosol models
tdm^v	upward diffuse transmittance model established with assuming that the single scattering albedo multiply forward scattering probability is 0.95 for j^{th} aerosol models
t_{ww}	$t_{ww}^s \times t_{ww}^v$
τ_r	Rayleigh optical thickness
$\eta(\lambda)$	forward scattering probability of aerosols
F_0	the extraterrestrial solar irradiance with considering sun-earth distance
f_0	extraterrestrial solar irradiance without considering sun-earth distance
C_{wc}	spectral correction function (Frouin et al., 1996) for whitecap reflectance
ws	wind speed at the sea level in m/s
ww	water vapor concentration in g/cm ²
C_{s-e}	sun-earth distance coefficient
c_p	the air pressure in mb
d_y	day of year

symbols	description
M_i	i^{th} aerosol model
w^{M_i}	weighting factor of M_i for aerosol reflectance fraction
f_{surf}	air-sea Fresnel reflectance ratio

Atmospheric correction is a primary process in satellite ocean color missions, in which surface water reflectance, ρ_{wn} , is extracted from the top of atmosphere (TOA). The quantity of ρ_{wn} is generally less than 10% of the TOA reflectance (ρ_{TOA}); thus, atmospheric reflectance and transmittance should be computed as accurately as possible. Ignoring the surface-reflectance from sun-glint and whitecaps, ρ_{TOA} at each wavelength (λ) can be described as follows:

$$\rho_{\text{TOA}} = \rho_r + \rho_a + \rho_{ra} + t_d^s t_d^v \rho_{\text{wn}} \quad (1)$$

where ρ_r is the Rayleigh multiple-scattering (molecular multiple-scattering) reflectance in the absence of aerosols, and $\rho_a + \rho_{ra}$ is the aerosol multiple-scattering reflectance in the presence of air molecules (denoted ρ_{am}). Terms t_d^s and t_d^v are the diffuse transmittances of the atmosphere from the sun to the sea surface and from the sea surface to the sensor, respectively.

Traditional atmospheric correction processes first compute $\rho_r(\lambda)$ within 1% error using a radiative transfer simulation [11–15]. Following subtraction of the Rayleigh reflectance contribution, $\rho_{\text{am}}(\text{NIR})$ can be estimated by assuming that the NIR water-leaving reflectance is negligible, i.e., $\rho_{\text{wn}}(\text{NIR}) = 0$, due to the relatively strong light absorption of the waterbody or so-called “black pixel” assumption. $\rho_{\text{am}}(\text{VIS})$ is then extrapolated from satellite-observed $\rho_{\text{am}}(\text{NIR})$ by inversion of the radiative transfer simulation.

2.5.2.1 Correction of gaseous absorption and white cap

Atmospheric correction algorithms generally start with correcting the gaseous absorption and white cap radiance emerged by wind stress at the sea level which can be predicted by meteorological information that are ozone, water vapor concentration, and wind speed as

$$L_{\text{TOA}}^{\text{cor}}(\lambda) = \frac{L_{\text{TOA}}(\lambda) - t d_r^v L_{\text{wc}}(\lambda)}{t_{\text{oz}}^s(\lambda) t_{\text{oz}}^v(\lambda)},$$

where $L_{\text{TOA}}(\lambda)$ is the TOA radiance observed by satellite at a wavelength λ , $L_{\text{TOA}}^{\text{cor}}$ is the TOA radiance after the correction of white cap and gaseous absorption, and $t d_r^v$ is the upward diffuse transmittance by air molecules from the sea surface to the sensor. Terms t_{oz}^s and t_{oz}^v are the downward gaseous transmittance by ozone from the sun to the sea surface and upward gaseous transmittance from the sea surface to the sensor, respectively. These ozone-absorptive transmittances t_{oz}^s and t_{oz}^v can be computed by quasi-analytic model as

$$t_{\text{oz}}^s(\lambda) = e^{\frac{-\tau_{\text{oz}}(\lambda)}{\cos(\theta_s)}},$$

$$t_{\text{oz}}^v(\lambda) = e^{\frac{-\tau_{\text{oz}}(\lambda)}{\cos(\theta_v)}},$$

where $\tau_{\text{oz}}(\lambda)$ is the optical thickness of ozone, θ_s is the solar zenith angle at the sea surface of the target point, and θ_v is the satellite zenith angle at the sea surface of the target point.

Whitecap reflectance can be simplified as a function of wind speed (Stramska and Petelski, 2003) and wavelength (Frouin et al., 1996). In this study, the effect of gas transfer efficiency by temperature is ignored due to its negligible contribution (Blanchard, 1971). It should be noted that spectral correction function (Frouin et al., 1996) for whitecap reflectance is adjusted for the GOCI-II SRF.

Meanwhile, the effect of water vapor absorption for correction of gaseous absorption in the GOCI algorithm has been ignored, because GOCI and GOCI-II bands are designed to avoid the absorption spectra. Recently revised investigation has showed that water vapor absorption

effect on GOCI bands 660, 745, and 865 nm for large zenith angles can be significant due to the absorption spectrum crossing in the out of band response range.

For the correction, 2nd order polynomial relationship between $\ln\{-\ln(wv)\}$ and $\ln\{(1/\cos \theta_s + 1/\cos \theta_v) \times wv\}$ is used contrariwise method for SeaWiFS, MODIS, and VIIRS uses linear relationship (Vermeulen, SeaWiFS Data Analysis System source code), then a simple estimating model for water vapor absorption (i.e., $t_{wv} = t_{wv}^s \times t_{wv}^v$) at the TOA level is developed using the radiative transfer simulation.

2.5.2.2 Solar irradiance normalization

The next step subsequent to the correction of gaseous absorption and whitecap correction is the solar irradiance normalization that is converting the TOA radiance into the TOA reflectance (ρ_{TOA}) as

$$\rho_{TOA}(\lambda) = \frac{\pi L_{TOA}^c(\lambda)}{F_0(\lambda) \cos(\theta_s)},$$

$$F_0(\lambda) = f_0(\lambda) \times c_{s-e}(d_y)$$

where θ_s is the solar zenith angle. The term f_0 is the extraterrestrial solar irradiance without considering sun-earth distance. Terms c_{s-e} and d_y are the sun-earth distance coefficient and day of year.

A full scene of GOCI-II is consisted by 12 sub-images (and is called as slots) that is obtained by step and stare method. Each slot is consisted by 8 band subsequently. Therefore, each band and each slot are obtained in different time, thus the time differences by introducing solar angles changes while compute the geometries should be considered.

After converting to the reflectance, the TOA reflectance can be divided by following contributions as

$$\rho_{TOA}(\lambda) = \rho_r(\lambda) + \rho_a(\lambda) + \rho_{ra}(\lambda) + tg^s(\lambda)tg^v(\lambda)\rho_g + td^s(\lambda)td^v(\lambda)\rho_{wn}(\lambda) \quad (2)$$

where ρ_r is the multiple scattering by air molecules (Rayleigh scattering) in the absence of aerosols, ρ_a is the multiple scattering by aerosols in the absence of air molecules, ρ_{ra} is the light interaction between air molecules and aerosols. Terms tg^s and tg^v are the total upward and downward direct transmittances including both the aerosols and the air molecules. Terms td^s and td^v are the total upward and downward diffuse transmittances including both the aerosols and the air molecules. The term ρ_{wn} is the water reflectance at sea surface in the absence of atmosphere. The term ρ_g is the sun-glint reflectance that is spectrally independent. However, the sun-glint has been omitted due to its relatively small contribution in GOCI-II sun-sensor geometries.

2.5.2.3 Rayleigh scattering (Molecular scattering)

Rayleigh scattering (ρ_r) takes the major quantitative part of the atmospheric reflectance (Wang, 2016). It can be horizontally fluctuated within a few percent by the air pressure changing, different geometries, and wind field distribution at the sea surface level. By the radiative transfer simulation, the Rayleigh component can be predicted less than one percent of error (Gordon et al., 1988; Gordon and Wang, 1992; Wang, 2002, 2016).

It is computationally intensive to directly calculate each pixel's Rayleigh reflectance. Thus, interpolated data from a pre-computed values from radiative transfer simulation is generally employed (Wang, 2003) for the Rayleigh correction. Four-dimensional lookup table is established for the most dominant parameters that are solar zenith angle, sensor zenith angle, relative azimuth angle, and wind. 6SV version 1 radiative transfer code and US Standard 62 atmospheric composition (without O₃ and H₂O) and profile model (McClatchey et al., 1971) are used for the simulation. The Rayleigh optical thickness correction is subsequently applied with considering the air-pressure variation (Wang, 2005) and the gravity effect (Ahn, 2016).

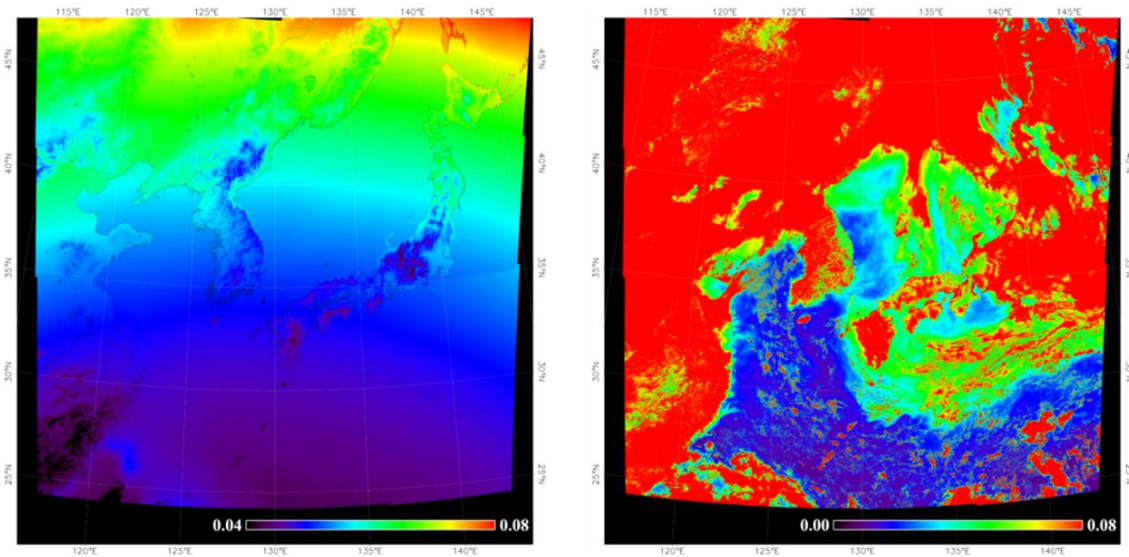


Figure 7. An example of Rayleigh reflectance correction (2013/08/13, 03:16 UTC). (Left) Rayleigh reflectance at 555 nm. (Right) Rayleigh corrected reflectance ($\rho_{TOA} - p_r$) at 555 nm.

Figure 7 is an example result of Rayleigh reflectance correction used for the GOCI standard atmospheric correction in 2013/08/ 13 03:16 UTC.

2.5.2.4 Cloud mask

Earlier studies, the cloud mask had accounted for only the high aerosol optical thickness (AOT) or clouds. In this cases, the simplest way of detecting clouds or high AOT is the use of a $\rho_{TOA}(NIR) - \rho_r(NIR)$ threshold, with the underlying assumption that relatively high absorbance of water at NIR wavelengths. However, since the ocean color remote sensing had been considered in turbid coastal waters, those masks over turbid waters should also be considered (Wang and Shi, 2006). For highly turbid waters, the ocean has a considerable reflectance by the backscattering of suspended sediment. Thus use a threshold of a band's reflectance over turbid waters can be failed. To separate turbid water and a high AOT, it is needed to use multiple bands. In this study, the scheme used two NIR bands' (745 nm and 865 nm) slope and threshold of $\rho_{TOA}(NIR) - \rho_r(NIR)$ value of the GOCI based on the theory of water's absorption always make two NIR wavelengths' slope higher than the aerosol's slope, and these threshold values were decided by a simple test based method. It can be approximated that reflectance ratio of ocean's two NIR bands' is always same (Ruddick et al., 2006) and give constraint that aerosol's slope range is around 0.9 to 1.2 from the assumption of aerosol types are only the oceanic or maritime. Selected pixels by threshold are applied to around 3x3 pixels due to the removal of sparkles that coming from the cloud's movement during band change. The result of this masking method is shown in the Figure 8.

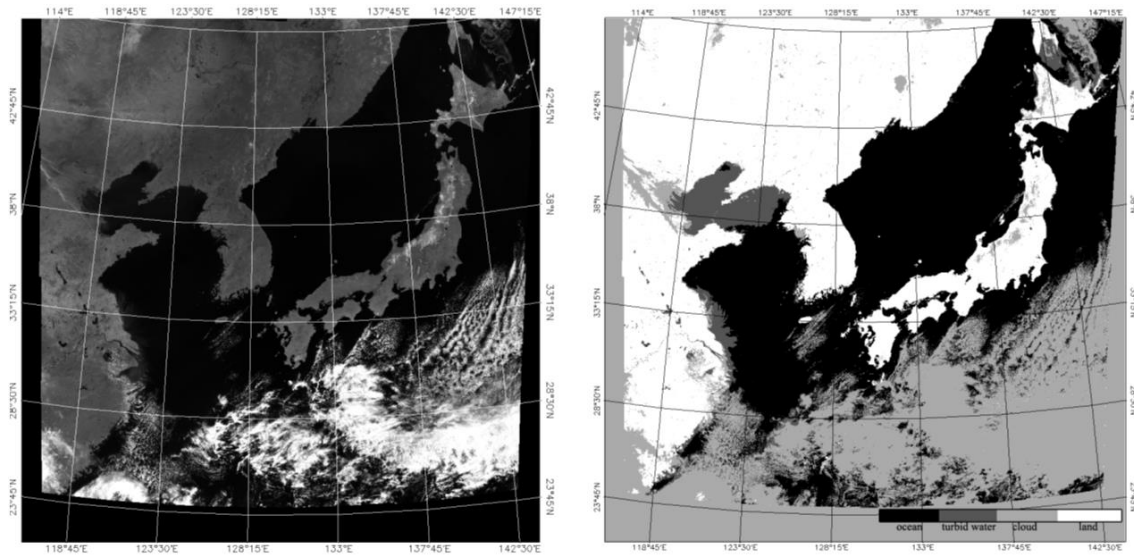


Figure 8. An example result of cloud masking for both clear and turbid waters (2011/04/05 03:16 UTC). Left image is Rayleigh corrected reflectance at 865 nm and right image is result of cloud screening.

2.5.2.5 Correction for the multiple scattering by aerosols - SRAMS

The SSE aerosol correction scheme (Gordon and Wang, 1994; Wang and Gordon, 1994) has been widely adopted for not only the GOCI and SeaWiFS but also the Moderate Resolution Imaging Spectroradiometer (MODIS), the Visible Infrared Imaging Radiometer Suite (VIIRS), and the Global Imager (GLI) (Toratani et al., 2007). However, the aerosol correction approach based on the SSE has several issues regarding the determination of appropriate aerosol models and their fraction (Ahn et al., 2016; Ahmad and Franz, 2016). In the process, the two most appropriate aerosol models and their mixture ratio (*i.e.*, weighting factor) are determined by comparison of the average SSE in NIR for all candidate aerosol models with assuming that multiple- to single-scattering conversion is not sensitive to aerosol models in NIR. It implies that all candidate models contribute to the model selection process although some of them are not really part of any cause. And the linear weighting factor computed in the single-scattering domain is directly used in the multiple-scattering domain to be used for extrapolation to the visible bands. Therefore, the estimated aerosol reflectance of the two models do not add up to the observed aerosol reflectance in NIR, and it may cause some residual errors. These irrelevant models can erroneously effect the mixture ratio of two appropriate models with inducing residual errors that will be amplified during the aerosol reflectance extrapolation from NIR to VIS wavelengths.

The atmospheric correction algorithm for GOCI and GOCI-II employs an alternative scheme that estimates reflectance fraction of the two models in the multiple-scattering domain directly, without going through the single-scattering domain to select appropriate aerosol models and the spectral extrapolation of their reflectance from NIR to VIS bands based on Ahn et al. (2016).

In the algorithm, following nine aerosol models based on the Shettle and Fenn (1979) is used as candidates, an oceanic model with relative humidity (RH) 99% (O99), maritime models with RH 50, 70, 90, and 95% (M50, M70, M90, and M95), coastal models with RH 50 and 70% (C50 and C70), tropospheric models with RH 50 and 80% (T50 and T80). From above models, O99 has the lowest spectral slope due to the largest particle size (a mixture of accumulate-mode mean radius 0.995 μm and coarse-mode mean radius 8.59 μm), and T50 has the highest spectral slope due to smallest particle size (mean radius 0.0275 μm).

2.5.2.6 Sun-glint correction

Sun-glint is the specular reflectance of sunlight at the roughened sea surface. The effect has been omitted for the GOCI atmospheric correction initially because the sun-glint effect is

relatively small for GOCI sun-sensor geometries. Nevertheless sun-glint effect in GOCI data is less than other polar orbit sensors, a small direct reflectance of sunlight by ocean surface can cause a considerable error in the southern part of GOCI imagery during summer season because the sun light itself is still relatively massive. Generally, the model of Cox & Munk (1954) is used to remove the sun-glint effect ρ_g .

Figure 9 is an example image of sun-glint reflectance for an GOCI scene in 2013/08/13 03:16 UTC.

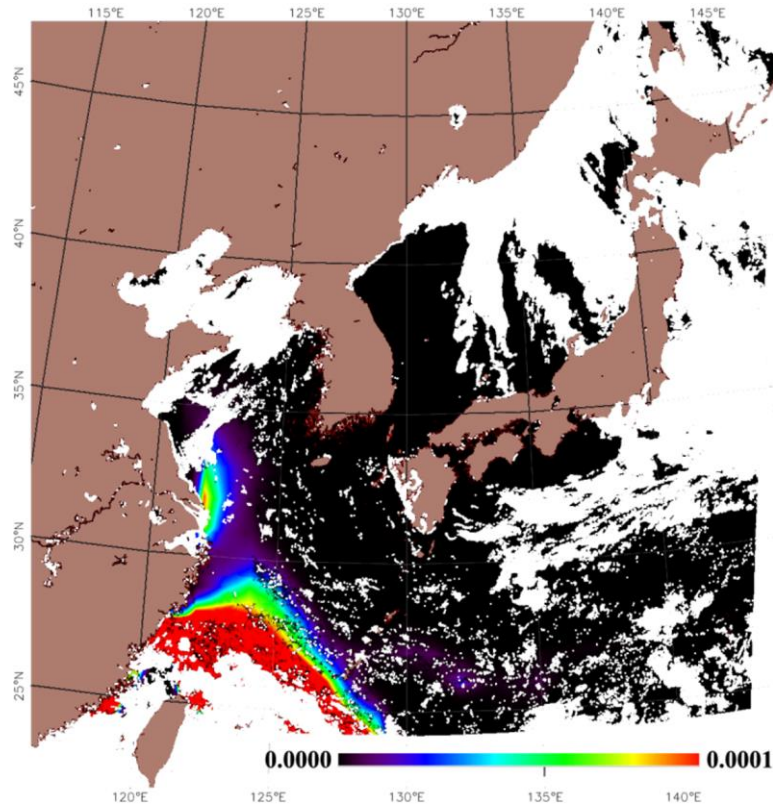


Figure 9. An example of sun-glint reflectance (2013/08/13 03:16 UTC) estimated by Cox & Munk (1954).

2.5.2.7 Correction for atmospheric diffuse transmittance

The early diffuse transmittances estimation in the atmospheric correction process had been estimated by a quasi-analytic model with an assumption of isotropic water-leaving radiance and single-scattering assumption (Gordon et al., 1983; Ahn et al., 2012). However, this assumption is no more stand for high zenith angles and high aerosol optical thickness, and then it starts to produce errors up to 15% (Antoine, 2010).

To simulate anisotropic water reflectance in diffuse transmittance estimation, Chl_a concentration is fixed as 0.7 mg/m^3 acceptable because in-water bidirectional effect by water constituents is relatively less sensitive compared to other terms (Park and Ruddick, 2005; Antoine, 2010) for large zenith angles. The Rayleigh diffuse transmittance is simply predicted by using RTE-organized LUT regarding solar zenith angles and viewing zenith angles.

For the diffuse transmittance of aerosols in the presence of air molecules ($td_a \times td_{ra}$: hereafter refer to as td_{am}), the GOCI atmospheric correction employed a similar method with Wang (1999) and Antoine (2010) that uses a relationship between an approximate model and result of radiative transfer simulation. The approximate diffuse transmittance models tdm^s and tdm^v are established with assuming that the single scattering albedo multiply forward scattering probability is 0.95.

2.5.2.8 NIR correction over turbid water

The GOCI and GOCI-II iteratively separate the water reflectance and aerosol reflectance in NIR wavelengths by using NIR water reflectance model. The NIR reflectance model uses spectral relationship between $\rho_{wn}(709)$ and the two $\rho_{wn}(745)$ and $\rho_{wn}(865)$ based on the Ruddick et al [2006].

The relationship model is empirically established from satellite-observed data by using nearest non-turbid water atmospheric correction (Hu et al., 2000).

2.5.2.9 Bidirectional effect correction

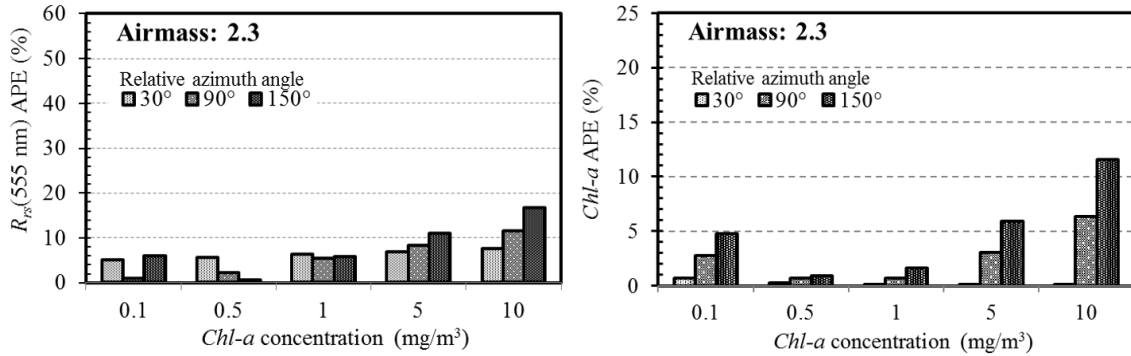


Figure 10. Impacts of bidirectional effect on level-2 product, $R_{rs}(555 \text{ nm})$ and Chl-a.

The bidirectional effect correction factor can be emerged by two major factors, one is the underwater contributions (f/Q) by volumetric phase function of suspended particles (e.g., phytoplankton, inorganic suspended matters, detritus, etc.) and the other is Fresnel transmittance ($\mathfrak{R}$) at the air-sea interface (i.e., $c_{BF} = \mathfrak{R} \cdot f/Q$). In the case of the lower zenith angle, the effects are dominated by the underwater constitutes. In contrast the higher zenith angle case, the directionality is more affected by the sea surface's Fresnel effect (Park and Ruddick, 2005).

The in-water bidirectional function f/Q is modeled with assuming that the living and non-living biogenic particles in the water column are the major regulating factor in f/Q fluctuations ($-f/Q$ is a function of chl_a concentration). The f/Q value is computed using a look-up table simulated (Monte Carlo method) and provided by Morel et al. (2005). The theoretical basis of their work was reported in Morel et al. (2002).

Figure 11 is an example of the BRDF correction result (c_{BF}) computed in the GOCI atmospheric correction (13th/Aug./2013 03:16 UTC).

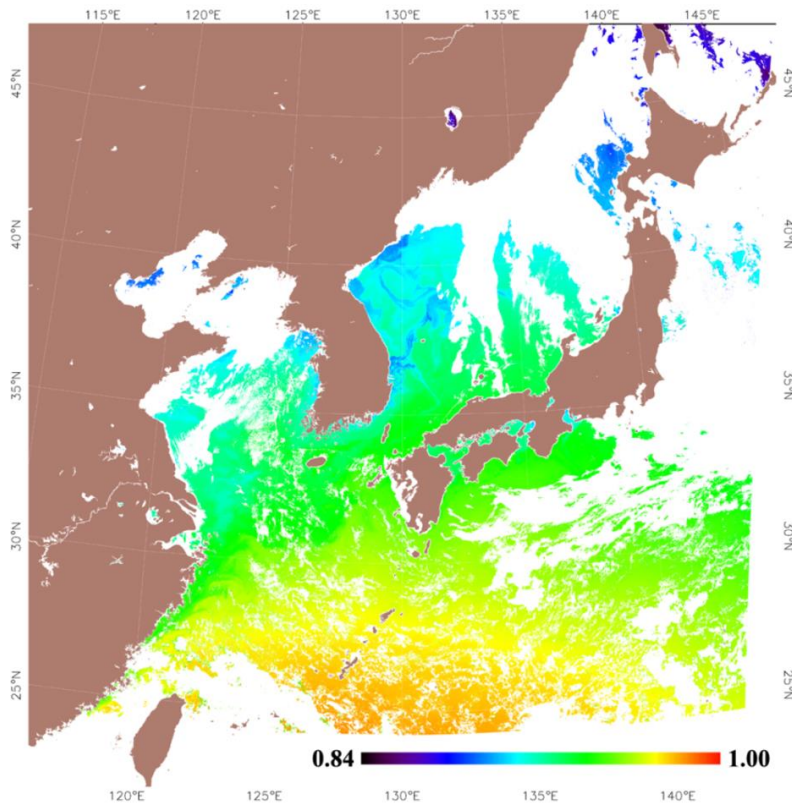


Figure 11. An example of BRDF correction factor (cBF) at 555 nm in the GOCI atmospheric correction process(2011/08/13, 3:16 UTC) (Morel et al., 2005; Wang, 2006; Ahn et al., 2015).

2.6 Algorithm Product Output

2.6.1 Output

- File format: NetCDF-CF
- Data form: 32bit float
- Dimension: 2780 pixels * 2780 pixels

(Data form will be changed to 16 bit in the future for better data service by reducing the data size)

2.6.1.1 RhoC (Rayleigh Corrected Reflectance)

Attribute	Precision	Value
standard_name	character	rayleigh_corrected_reflectance_[wavelength]
long_name	character	Rayleigh corrected reflectance at [wavelength] nm from TOA, KOSC standard algorithm
_FillValue	float	-999.0

Attribute	Precision	Value
units	character	-
valid_min	float	0
valid_max	float	1
scale_factor	float	1.0
add_offset	float	0.0

2.6.1.2 R_{rs} (Remote Sensing Reflectance)

Attribute	Precision	Value
standard_name	character	remote_sensing_reflectance_[wavelength]
long_name	character	Remote sensing reflectance at [wavelength] nm, KOSC standard algorithm
_FillValue	float	-999.0
units	character	sr-1
valid_min	float	-0.01
valid_max	float	0.1
scale_factor	float	1.0
add_offset	float	0.0

2.6.2 Flag

flag_masks (Bit)	flag_values	flag_meanings	Descriptions
1	1	Cloud_or_Ice	Cloud or ice
2	2	Land	Land
3	4	AC_Fail	Atmospheric correction failure
4	8	Extrm_Turbid	Extremely turbid water ($R_{rs} 660 > 0.020$)
5	16	High_SZA	High solar zenith angle ($SZA > 70$)
6	32	High_VZA	High satellite zenith angle ($VZA > 55$)

flag_masks (Bit)	flag_values	flag_meanings	Descriptions
7	64	Anc_Miss	Missing ancillary data both near-realtime and averaged trend
8	128	Ozone_Warn	Ozone concentration is +-50DU of annual average cycle
9	256	High_WS	Wind speed >= 10 m/s
10	512	High_AOT	High aerosol load, Rho_a(865) + Rho_ra(865) > 80% of threshold
11	1024	High_Aero_Ratio	Aerosol reflectance takes over 300% of water reflectance at any bands
12	2048	Eps_Warn	MS Epsilon(865, 745) is out of range
13	4096	Neg_Lw	Negative Lw in any band
14	8192	Low_Lw_555	Low Lw at 555 nm
15	16384	Neg_Lw_NIR	Negative Lw at the NIR band
16	32768	Neg_RhoC_NIR	Negative RhoC at the NIR band
17	65536	Case2_Warn	Case2 water warning
18	131072	High_Sunlint	High Sunlint
19	262144	Cloud_Edge	Cloud Edge
20	524288	Turbid_Water	Turbid Water
21	1048576	Bright_Adj	Bright pixel (cloud or land) or adjacency pixel (stray light)
22	2097152	Contm_Warn	High possibility of unavailable pixel contamination (masked pixel within 3x3 bound)
23	4194304	Slot_Warn	Slot edge contamination
24	8388608	Chl_Fail	Chl-a is not calculable
25	16777216	Chl_Warn	Chl-a is out of range
26	33554432	Rrs_Fail	Rrs is not calculable (Spectrum shape is wrong)
27	67108864	Rrs_Warn	Rrs is out of range
28	134217728	Spectral_Dubious	Rrs Spectrum shape is statistically distrustful (H.Fukushima)

flag_masks (Bit)	flag_values	flag_meanings	Descriptions
29	268435456	Spectral_Wrong	Rrs (or nLw) Spectrum shape is wrong
30	536870912	Inhomo_Warn	Homogeneous test fail
31	1073741824	Floating_Algae	Floating Algae
32	2147483648	Abs_Aerosol	Absorbing aerosol warning

3 TEST DATA SETS AND OUTPUT

3.1 Simulated / Proxy Input Datasets

Two types of validation are conducted for the algorithm. The validation on the code implementation (test on whether the code is implemented to the GOCI-II data processing system identically to that developed by developers) is rather a technical verification process. Validation on the algorithm performance will be main focus of this section.

Test data used for the validation of G2AC algorithm are categorized into three types; 1) in-situ data, 2) simulation data, and 3) satellite data.

1) In-situ datasets

- A. In-situ data are often regarded as the reference data with little errors, but are not free of error due to observation uncertainty.
- B. In-situ data from field campaigns conducted by KIOST and AERONET-OC data operated by NASA data were used.

2) Simulation datasets

- A. L_{TOA} and R_{rs} data simulated with radiative transfer code such as AccuRT or 6SV

3) Satellite data

- A. Other satellite R_{rs} produced by VIIRS, MODIS, and OLCI.

3.1.1 In situ radiometric data measured from shipboard

In situ radiometric dataset for G2AC validation are not collected yet.

3.1.2 AERONET-OC radiometric data

At the leodo station (Shim et al., 2004), there is no level 2.0 AERONET-OC data that the quality is fully assured (Zibordi et al., 2006; Zibordi et al., 2009). Moreover, Moon et al. (2012) requires IOP pairs to build R_{rs} optical closure, however, AERONET-OC does not measure them. Therefore, a new data screening approach is applied that uses a different method from Moon et al. (2012). To set optical boundaries, a series of empirical relationships of R_{rs} spectra are simulated by HYDROLIGHT (Mobley and Sundman, 2008).

In this study, various R_{rs} spectra are computed using the following input parameters:

- Pure water IOPs are taken from Smith and Baker (1981), Pope and Fry (1997), and Kou et al.(1993).
- We set chl_a concentration range as 0.1~30.0 mg/m³. Then, its IOP are taken from Morel (1988), and Loisel and Morel (1998). Specific absorption of chl_a in 740~880 nm is assumed to be zero.
- CDOM absorption range at 440 nm considered is 0.1~0.3 m⁻¹.
- Suspended sediment concentration range is 0.1~1000.0 g/m³. From Ahn (1990), IOP of 4 mineral types, namely red clay (RC), yellow clay (YC), calcareous sand (CS), and brown earth (BE), were considered. Each absorption and backscattering coefficient spectra were extrapolated over the NIR range (Mobley and Sundman, 2008).
- Inelastic scattering effects (chl_a fluorescence, CDOM fluorescence, and Raman scattering) are considered.
- Wavelength range of the simulation considered is 400~880 nm per 5 nm bandwidth. Then convert into AERONET-OC bands.

Finally, the following spectral relationships are developed; between $R_{rs}(443\text{ nm})/R_{rs}(555\text{ nm})$ and $R_{rs}(668\text{ nm})$ [Figure 12(a)], between $R_{rs}(412\text{ nm})/R_{rs}(443\text{ nm})$ and $R_{rs}(668\text{ nm})$ [Figure

12(b)], between $R_{rs}(443 \text{ nm})/R_{rs}(490 \text{ nm})$ and $R_{rs}(668 \text{ nm})$ [Figure 12(c)], between $R_{rs}(668 \text{ nm})/R_{rs}(555 \text{ nm})$ and $R_{rs}(668 \text{ nm})$ [Figure 12(d)], and between $R_{rs}(869 \text{ nm})/R_{rs}(668 \text{ nm})$ and $R_{rs}(668 \text{ nm})$ [Figure 12(e)]. Then some data are excluded which exhibited spectral relationships out of boundary conditions (blue-dashed closures) [Figure 12]. Except the relationship between $R_{rs}(869 \text{ nm})/R_{rs}(668 \text{ nm})$ and $R_{rs}(668 \text{ nm})$ [Figure 12(e)], those boundaries do not consider $R_{rs}(668)$ less than 0.0045 because these relationships showed high variations in this reflectance range.

It should be noted that the verification of IOP models is beyond the scope of this study. Moreover, those IOP models have not been developed for the GOCI area, hence they would not be consistent with the underwater optical environments of the calibration sites. Moreover, the extrapolated IOPs in the NIR may lead to unrealistic relationships between $R_{rs}(869 \text{ nm})/R_{rs}(668 \text{ nm})$ and $R_{rs}(668 \text{ nm})$. For these reasons, further investigations regarding IOPs at the study sites are needed for future improvement. Because of a lack of IOP verifications, the method excluded 60 unrealistic spectra as shown in Figure 13.

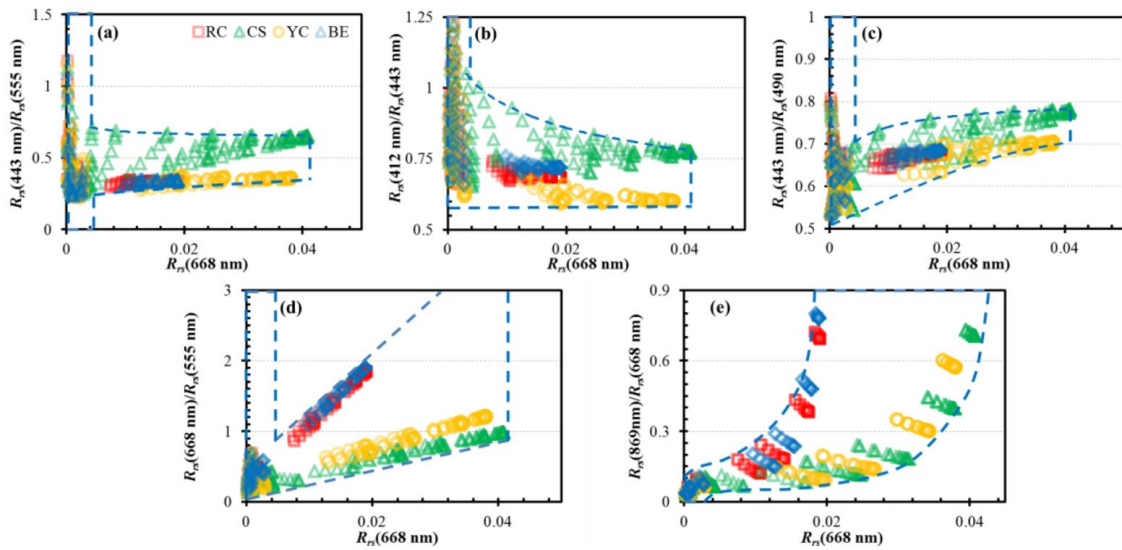


Figure 12. R_{rs} spectral relationships obtained through HYDROLIGHT simulations (Ahn et al., 2015). For this simulation, the range of Chla concentration varied from 0.1~30 mg m⁻³, CDOM absorption at 440 nm from 0.1~0.3 m⁻¹, and suspended sediment concentration from 0.1~1000 g m⁻³.

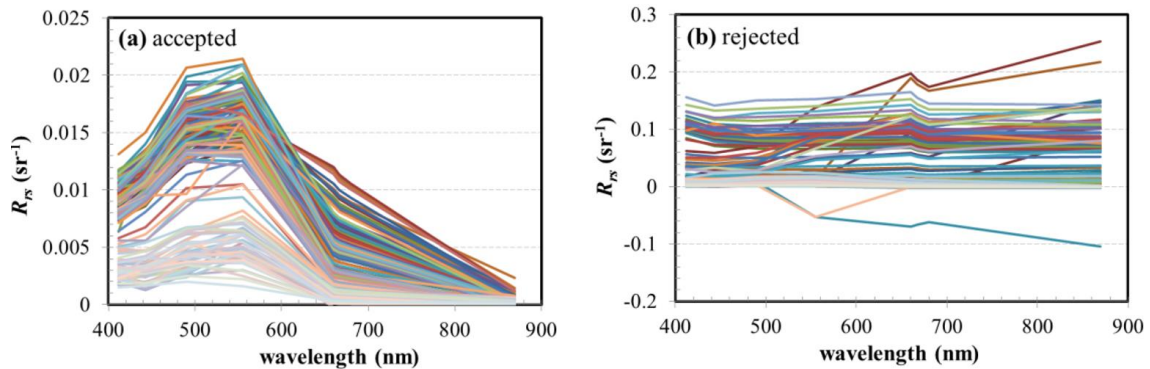


Figure 13. Quality control adopted for the AERONET-OC data (Ahn et al., 2015). (a) R_{rs} spectra accepted by the quality control screening criteria, and (b) R_{rs} spectra rejected by the scheme because of spurious outliers.

3.1.3 Simulation dataset

To test the reliability of the atmospheric correction using the SRAMS approach, various pseudo-TOA reflectances were generated by Eq.(2) using a radiative transfer simulation (Vermote et al., 2006) for atmospheric components and an biogenic ocean color model (Morel and Maritorena, 2001) for water reflectance (Figure 14). Six aerosol models based on Shettle and Fenn (1979) but not identical to 9 candidate models –Maritime, Coastal, and Tropospheric models with different RH (i.e., M98, M80, C90, C80, T90, and T50) were used to compute atmospheric reflectance and transmittance for observation geometries: $\theta_s = 0^\circ, 25^\circ, 50^\circ$, and 75° , $\theta_v = 0^\circ, 20^\circ, 40^\circ$, and 60° , and ϕ_{sv} are from 0° to 180° with 15° gap.

Water reflectances were computed (Figure 14) for chl_a concentrations 0.03, 0.1, 0.3, and 1.0 mg/m^3 without inorganic particulate matter. It should be noted that case-2 water reflectance is ignored for the simulation because this study focuses on only the aerosol reflectance retrieval.

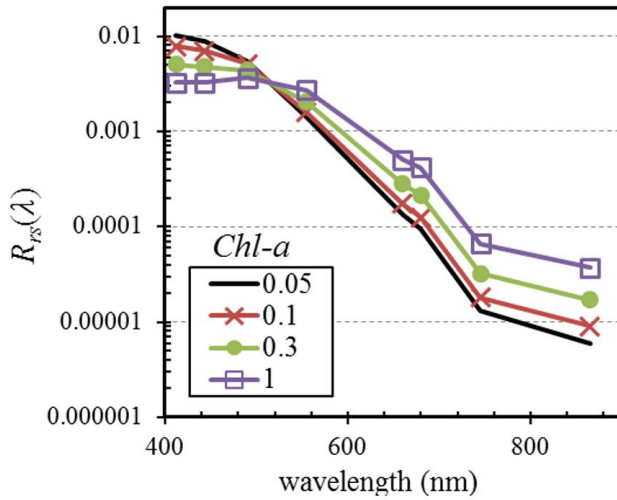


Figure 14 Remote-sensing reflectance for Chla concentrations 0.03, 0.1, 0.3, and 1.0 mg/m^3 without inorganic particulate matter by using biogenic ocean color model (Morel and Maritorena, 2001)

3.2 Output from Simulated Datasets

3.2.1 Accuracy Estimates

Evaluation is analyzed by using the statistical indices such as the absolute percentage error (APE) (i.e. absolute percentage distribution: APD) and root mean square error (RMSE) defined as follows:

$$APE = \frac{100}{K} \sum_{n=1}^K \left(\frac{|V_T^n - V_E^n|}{V_T^n} \right),$$

$$RMSE = \sqrt{\frac{\sum_{n=1}^K (v_n^t - v_n^e)^2}{K}},$$

where K is the total number of match-up pairs, and v_n^t and v_n^e are the true and derived values of n^{th} match-up entry, respectively.

4 PRACTICAL CONSIDERATIONS

4.1 Numerical Computation Considerations

Processing of the G2AC algorithm requires a PC/Workstation that is equipped with at least 8GB RAM. The required processing time is within 15 minutes in ADE. ADE is equipped with dodeca-core 2.5 GHz CPU and K80 GPU.

4.2 Algorithm Validations

The following guideline is provided for the validation of the G2AC products.

- (1) G2AC algorithm computes R_{rs} spectra in the final stage, where the uncertainty in the intermediate steps are accumulated. ...
- (2) Quality of each R_{rs} image can be validated by comparison with other satellites such as MODIS and VIIRS. Major target areas include North-west Pacific Ocean, the East Sea, etc.
- (3) The overall performance of the G2AC algorithm can be evaluated by the comparison with the *in-situ* R_{rs} measurements. *In-situ* R_{rs} data are available from field campaigns conducted by KIOST/KOSC, and also from NASA AERONET-OC and SeaBASS.

5 ASSUMPTIONS AND LIMITATIONS

This section provides explanation on the limitations in the system and the uncertainty of the input data.

5.1 Performance

The assumptions made in the development process is as follows.

1. Meteorological reprocessing data and climatology data has to be secured for hourly data processing.
2. GOCI-II Level 1B geometric/radiometric calibration must not have errors.
3. LUT input range need to satisfy the GOCI-II acquisition scan-geometry range.

5.2 Assumed Sensor Performance

The atmospheric correction algorithm requires 99.5% accuracy of sensor radiometric calibration. However, the general ocean color sensors' radiometric calibration uncertainties are larger than 3%. Therefore, additional system vicarious calibration will be applied.

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7 APPENDIX

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